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Aim of the experiment

Design and Test 2-bit Comparator

Objective

To verify 2 bit comparator using universal logic gate.

Apparatus Required

SL NO	Name of component	Specification	Quantity
1.	Trainer kit (analog and digital)	XPO-ANADIG1	1
2.	Bread board	—	1
3.	AND Gate (IC-74LS11)	Triple 3-input AND Gate	1
4.	OR Gate (IC 74LS32)	Quad 2 input OR Gate	1
5.	NOT Gate (IC 74LS 04)	Hex inverter	1
6.	XOR Gate (IC 74LS 86)	Quad 2 input XOR Gate	1
7	Connecting wires	—	As per Required

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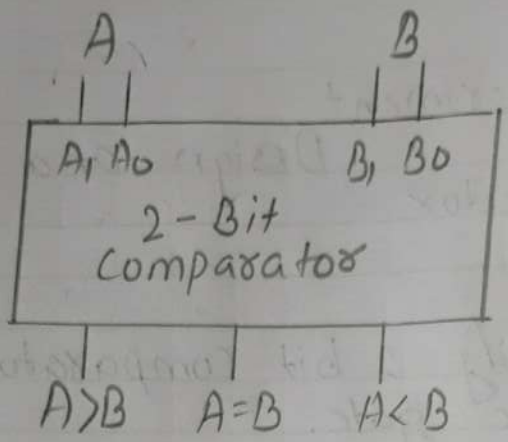


Fig 7.1: Block Diagram of 2 Bit Comparator

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Theory

A 2-bit Comparator is a combinational logic circuit used to compare two bit binary number $A = A_1A_0$ and $B = B_1B_0$. It determines whether A is less than, equal to, or greater than B.

The circuit has two 2-bit inputs and three output : $A < B$, $A = B$, and $A > B$. At a time only one output becomes High depending on the comparison between the two binary number

Truth Table :-

Table 7.1 :-

Input				Output		
A_1	A_0	B_1	B_0	$A<B$	$A=B$	$A>B$
0	0	0	0	0	1	0
0	0	0	1	1	0	0
0	0	1	0	1	0	0
0	0	1	1	1	0	0
0	1	0	0	0	0	1
0	1	0	1	0	1	0
0	1	1	0	1	0	0
0	1	1	1	1	0	0
1	0	0	0	0	0	1
1	0	0	1	0	0	1
1	0	1	0	0	1	0

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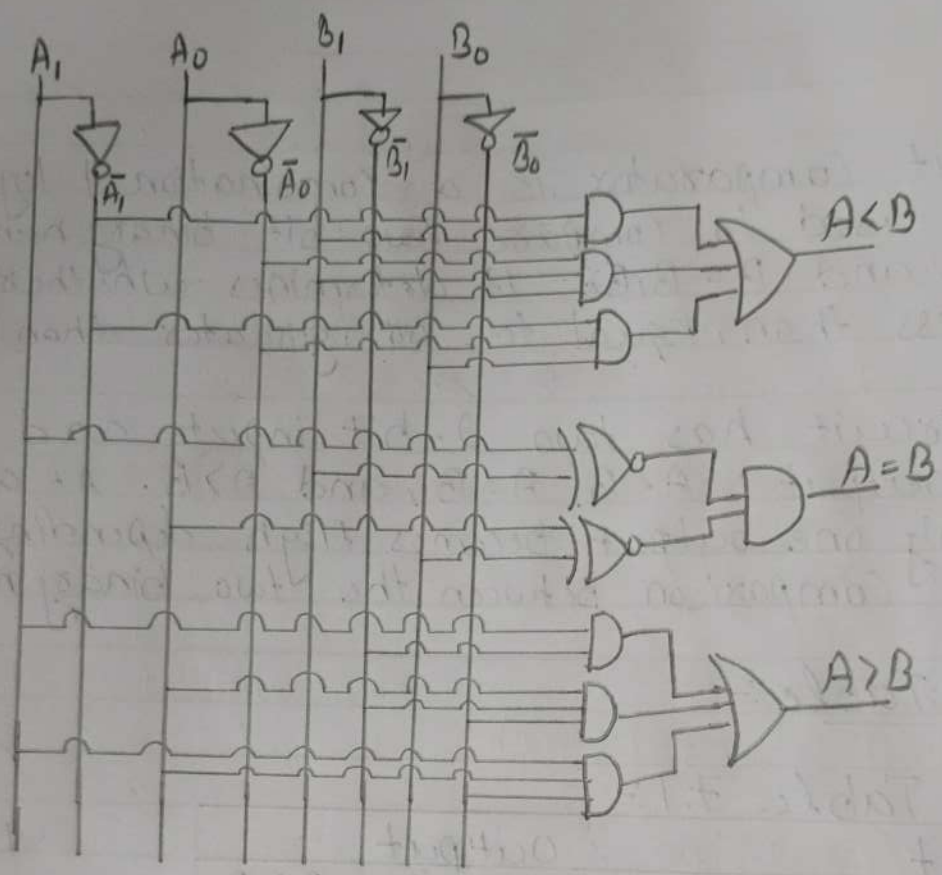


Fig 7.2: Logic circuit digram of 2-Bit comparator

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Table 7.1

Input				Output		
A_1	A_0	B_1	B_0	$A<B$	$A=B$	$A>B$
1	0	1	1	1	0	0
1	1	0	0	0	0	1
1	1	0	1	0	0	1
1	1	1	0	0	0	1
1	1	1	1	0	1	0

The Boolean expressions for the outputs are

1. For $A<B$

$$(A<B) = \Sigma m(1,2,3,6,7,11)$$

A_1A_0 \ B_1B_0	$\bar{B}_1\bar{B}_0$ 00	$\bar{B}_1B_0$ 01	B_1B_0 11	$B_1\bar{B}_0$ 10
$\bar{A}_1\bar{A}_0$ 00	0	1 2	1 3	1 2
$\bar{A}_1A_0$ 01	4	5	1 7	1 6
A_1A_0 11	12	13	15	14
$\bar{A}_1A_0$ 10	8	9	1 11	10

$$(A<B) = \bar{A}_1B_1 + \bar{A}_1\bar{A}_0B_0 + \bar{A}_1B_1B_0$$

2. For $A=B$

$$(A=B) = \Sigma m(0,5,10,15)$$

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A_1A_0 \ B_1B_0	$\bar{B}_1\bar{B}_0$ 00	$\bar{B}_1B_0$ 01	B_1B_0 11	$B_1\bar{B}_0$ 10
$\bar{A}_1\bar{A}_0$ 00	(1) 0	1	3	2
$\bar{A}_1A_0$ 01	4	(1) 5	7	6
A_1A_0 11	12	13	(1) 15	14
$A_1\bar{A}_0$ 10	8	9	11	(1) 10

$$(A = B) = (A_1 \odot B_1)(A_0 \odot B_0)$$

3. For $A > B$

$$(A > B) = \Sigma m(4, 8, 9, 12, 13, 14)$$

A_1A_0 \ B_1B_0	$\bar{B}_1\bar{B}_0$ 00	$\bar{B}_1B_0$ 01	B_1B_0 11	$B_1\bar{B}_0$ 10
$\bar{A}_1\bar{A}_0$ 00	0	1	3	2
$\bar{A}_1A_0$ 01	(1) 4	5	7	6
A_1A_0 11	(1) 12	(1) 13	15	(1) 14
$A_1\bar{A}_0$ 10	(1) 8	(1) 9	11	10

$$(A > B) = A_1\bar{B}_1 + A_1A_0\bar{B}_0 + A_0\bar{B}_1\bar{B}_0$$

Procedure

1. Identify the required ICs and collect the necessary connecting wires and digital trainer kit.

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7. Study the Boolean expression for the outputs $A<B$, $A=B$, and $A>B$ obtained using K-map simplification

3. Implement the simplified Boolean expression using the required universal logic gates

4. Connect the inputs A_1, A_0, B_1 and B_0 to the respective logic gates circuits.

5. Apply all possible combination of input value to the circuit

6. Observe the output corresponding to $A<B$, $A=B$, and $A>B$ for each input combination

7. Record the observed outputs in the truth table.

8. Compare the observed outputs with the theoretical truth table to verify the correct operation of the 2-bit Comparator

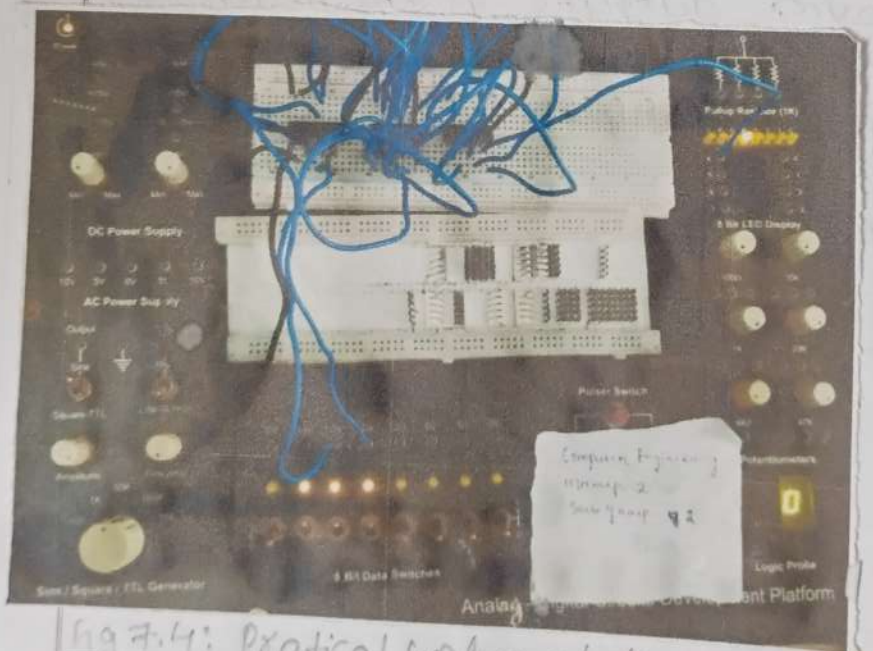
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Input				Output
$A_1(D_7)$	$A_0(D_6)$	$B_1(D_5)$	$B_0(D_4)$	$Y = A > B$ (4)
ON	OFF	OFF	OFF	Glowing

(ON = 1, OFF = 0)

Fig 7.3: Pratical Implementation of A > B for 2-Bit Comparator



Input				Output
A_1 (D_7)	A_0 (D_6)	B_1 (D_5)	B_0 (D_4)	$Y = A < B$ (L1)
OFF	ON	ON	ON	Glowing

(ON = 1, OFF = 0)

Fig 7.4: Pratical implimentation of A < B for 2-bit Comparato

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Observation Table

1. For A>B

Observation Table 1:-

Input				Output	
A_1 (D_7)	A_0 (D_6)	B_1 (D_5)	B_0 (D_4)	Y= A>B (4)	Remark
OFF	ON	OFF	OFF	Glowing	
ON	OFF	OFF	OFF	Glowing	The practical output matches the theoritical output
ON	OFF	OFF	ON	Glowing	
ON	ON	OFF	OFF	Glowing	
ON	ON	OFF	ON	Glowing	
ON	ON	ON	OFF	Glowing	

(ON=1, OFF=0)

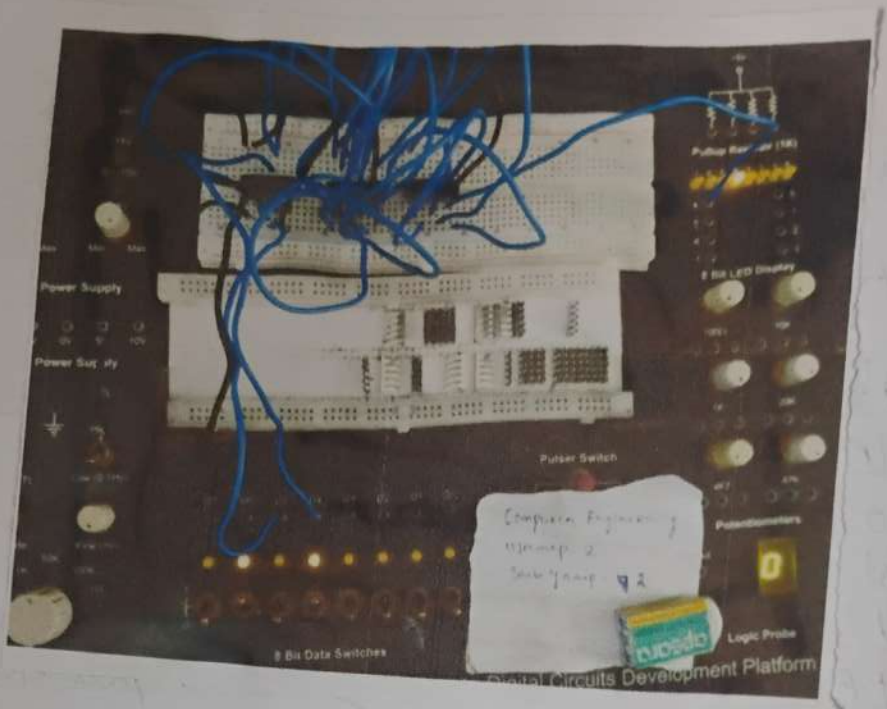
2. For A<B

observation Table 2:-

Input				Output	
A_1 (D_7)	A_0 (D_6)	B_1 (D_5)	B_0 (D_4)	Y=A<B (4)	Remark
OFF	OFF	OFF	ON	Glowing	
OFF	OFF	ON	OFF	Glowing	The pratical output matches the theoritical output
OFF	OFF	ON	ON	Glowing	
OFF	ON	ON	OFF	Glowing	
OFF	ON	ON	ON	Glowing	
ON	OFF	ON	ON	Glowing	

(ON=1, OFF=0)

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Input				Output
A_1 (D_7)	A_0 (D_6)	B_1 (D_5)	B_0 (D_4)	$Y=(A=B)$ (L1)
OFF	ON	OFF	ON	Glowing

(ON=1, OFF=0)

Fig 7.5: Pratical Implementation of A=B for 2-bit Comparator.

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3. For A = B

Observation Table 3:-

Input				output	
A_1	A_0	B_1	B_0	Y=(A=B)	Remark
(D_7)	(D_6)	(D_5)	(D_4)	(4)	
OFF	OFF	OFF	OFF	Glowing	
OFF	ON	OFF	ON	Glowing	The pratical output
ON	OFF	ON	OFF	Glowing	matches theoritical output
ON	ON	ON	ON	Glowing	

(ON = 1, OFF = 0)

Conclusion

Designing and testing 2-bit comparator using universal logic gates, implementing the the Boolean expression for A<B, A=B and A<B and verifying the correct operation of the circuit by comparing the observed outputs with the theoritical truth table for all possible input combinations.

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